# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I* S. A. Adnan (197226008 4) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

| Criteria | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | 1.75 | 1.75 | 1.75 | 1.75 |
| Explanation | 25% | 1.75 | 1.75 | 1.75 | 1.75 |
| Application | 20% | 1.4 | 1.4 | 1.4 | 1.4 |
| Organization | 15% | 1.05 | 1.05 | 1.05 | 1.05 |
| Accuracy | 10% | 0.7 | 0.7 | 0.7 | 0.7 |
| Clarity | 5% | 0.35 | 0.35 | 0.35 | 0.35 |

**Total Marks** 28 /40

***Signature of the Faculty***

## Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

**9.**A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

**10.**A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

**RUBRICS FOR CALCULATION**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25% | Demonstrates a thorough, accurate understanding of concepts | Good understanding with minor gaps | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation | 25% | Detailed, well-explained answers with relevant examples | Clear explanation with adequate details | Limited explanation; lacks depth | Poor or unclear explanation |

| | | | | | |
|---|---|---|---|---|---|
| Application | 20% | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors | Limited application with weak analysis | No application or incorrect analysis |
| Organization | 15% | Well-structured answers with logical flow and coherence | Organized with minor issues in flow | Basic structure, lacks clarity | Poor organization, incoherent answers |
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps |
| Clarity | 5% | Clear, neat, professional presentation | Understandable with minor issues | Limited clarity, readability issues | Poorly written, difficult to understand |

A company needs to transfer a 150 MB design file from a client computer to remote server over a network with a bandwidth of 50 Mbps. The transport layer divides the file into 1500-byte segments, while the Data link layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data link layer, and estimate the total transmission time (ignoring propagation delay) Explain how the transport and Data link layers ensure reliable communication.

Ans:- Total number of segments

$$\text{Segments} = \frac{150 \times 10^6}{1500}$$

$$= 100.000 \text{ segments}$$

Total number of frames :-

Assuming one segment is encapsulated into one frame

Total Data link layer overhead

$$= 100,000 \times 40 = 4,000,000 \text{ bytes}$$

$$= 4 \text{ MB}$$

Total transmission time

Total data transmitted, including overhead

$$150 + 4 = 154 \text{ MB}$$

Divide by bandwidth

$$T = \frac{154 \times 10^6 \times 8}{50 \times 10^6} = 24.64 \text{ seconds}$$

Transport layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans :- Number of transmission windows

$$\text{Windows} = \frac{48}{6} = 8 \text{ windows}$$

Total no. of retransmissions

One frame is lost in every windows and must be retransmitted.

Retransmission = 8 x 1

8 transmissions

Data transmitted per window

6 x 1024 = 6144 bytes

Explaination :- flow control prevents a fast sender from overwhelming a slow receiver. The sliding-window mechanism allows multiple frames to be sent before acknowledgments are received, improving bandwidth utilization and reducing waiting time. Lost frames are retransmitted according to the protocol.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network layer ensures Sucessful delivery of fragmented packets to the destination.

Sol:- Maximum data per fragment

$1500 - 20 = 1480$ bytes.

The orignal payload is $12000 - 20 = 11980$ bytes

Total no.of fragments

$$\text{fragments} = \frac{11980}{1480} = 9 \text{ fragments.}$$

Total header overhead

Overhead $= 9 \times 20 = 180$ bytes

Total transmitted size

$11980 + 180 =$ **12160 bytes**

Explaination :- The network layer divides a large IP Packet into smaller fragments to fit the MTU of the network. Each fragment contains its own IP header, including identification and fragmentation information. The destination uses this information to reassemble the fragments are missing, higher-layer protection such as TCP may help recover the lost data.

Transport layer protocol uses a sliding window size of 6 frames. Each frames, calculate the number 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be transmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Ans:. Number of transmission windows

$$\text{Windows} = \frac{48}{6}$$

$$= 8 \text{ windows}$$

Total number of retransmissions

One frame lost in every windows

$$\text{Retransmissions} = 8 \times 1$$

8 transmissions.

Data carried by each window.

$$6 \times 1024 = 6144 \text{ bytes}$$

Explaination :- flow control regulates the rate of data transmission so that the receiver can process incomming framing without being overwhelmed. The sliding-window method allows the sender to transmit several frames before waiting for acknowledgments.

This improves network efficiency, reduces idle time, and supports retransmission of lost frames to help ensure reliable communication.

# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME:** CSA07 **COURSE CODE:** Computer Networks

**COURSE OUTCOME COVERED**

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

**CONCEPT MAPPING**

*Code of Conduct:*

I Shaik Abdul Adnan (192260084) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

| Criteria | Weightage (%) | Q1 10marks | Q2 10marks | Q3 10marks |
|---|---|---|---|---|
| **Concept Identification** | 20% | 1.4 | 1.4 | 1.4 |
| **Concept Mapping Structure** | 25% | 1.75 | 1.75 | 1.75 |
| **Linking & Relationships** | 20% | 1.4 | 1.4 | 1.4 |
| **Application & Analysis** | 20% | 1.4 | 1.4 | 1.4 |
| **Explanation & Clarity** | 15% | 1.05 | 1.05 | 1.05 |

**Total Marks** 21/30 ***Signature of the Faculty***

**CONCEPT MAPPING QUESTIONS**

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖
Analyze the problem and explain how the concept map helps in troubleshooting.

**RUBRICS FOR CALCULATION**

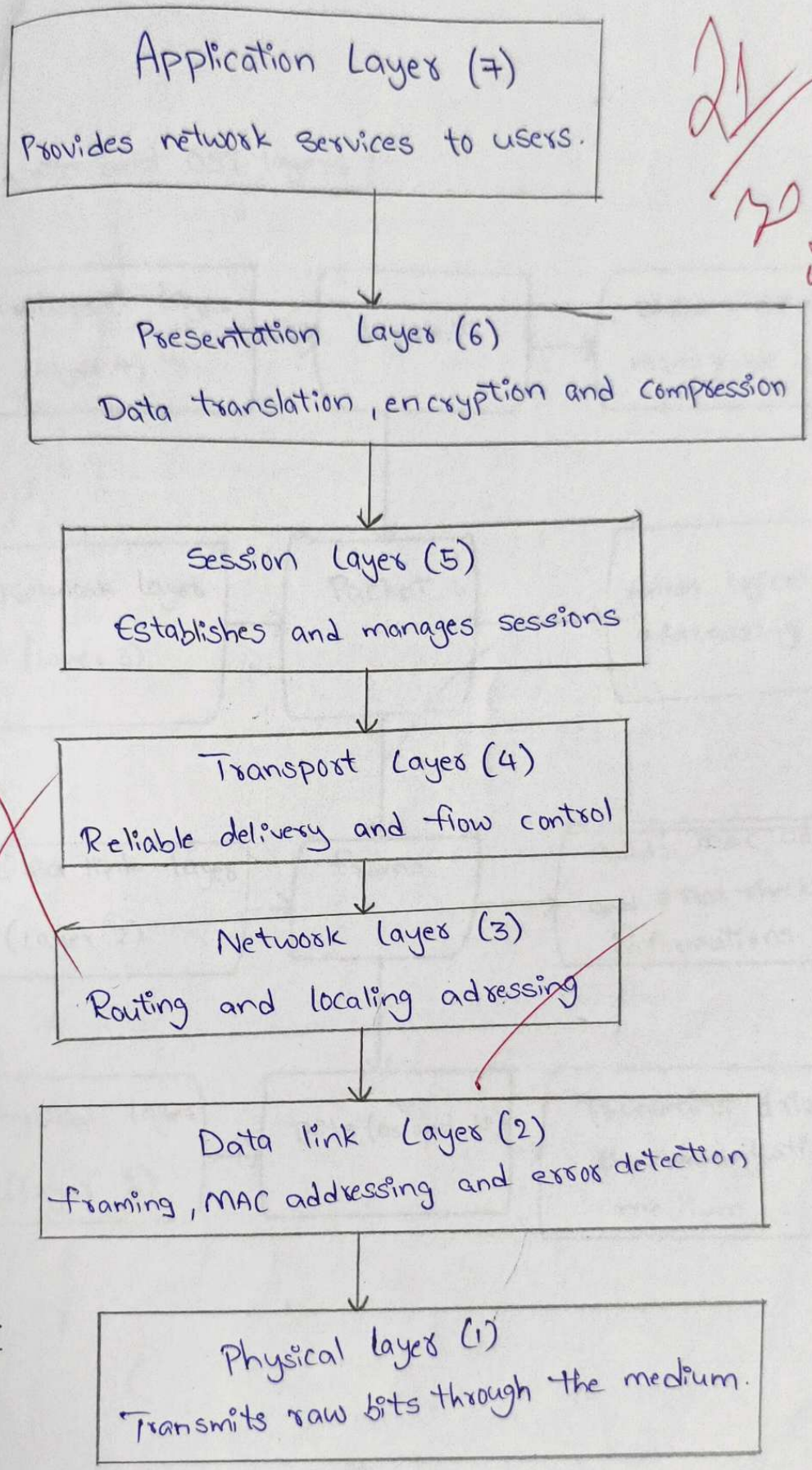
| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Concept Identification** | 20% | Accurately identifies all relevant OSI layers, concepts, and relationships | Identifies most concepts with minor omissions | Identifies limited concepts; some inaccuracies | Unable to identify key concepts correctly |
| **Concept Mapping Structure** | 25% | Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections | Concept map is mostly clear with minor structural issues | Concept map lacks clarity, organization, or proper flow | No clear structure or incorrect mapping |
| **Linking & Relationships** | 20% | Clearly establishes correct relationships between layers, functions, and processes | Relationships are mostly correct with minor errors | Limited or partially correct relationships | Incorrect or missing relationships |

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| **Application & Analysis** | 20% | Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning | The application is correct but lacks depth in analysis | Basic application with weak explanation | Unable to apply concepts correctly |
| **Explanation & Clarity** | 15% | Provides a clear, concise, and well-structured explanation supporting the concept map | Explanation is understandable with minor gaps | Explanation is unclear or incomplete | No clear or meaningful explanation |

AT2 - C01

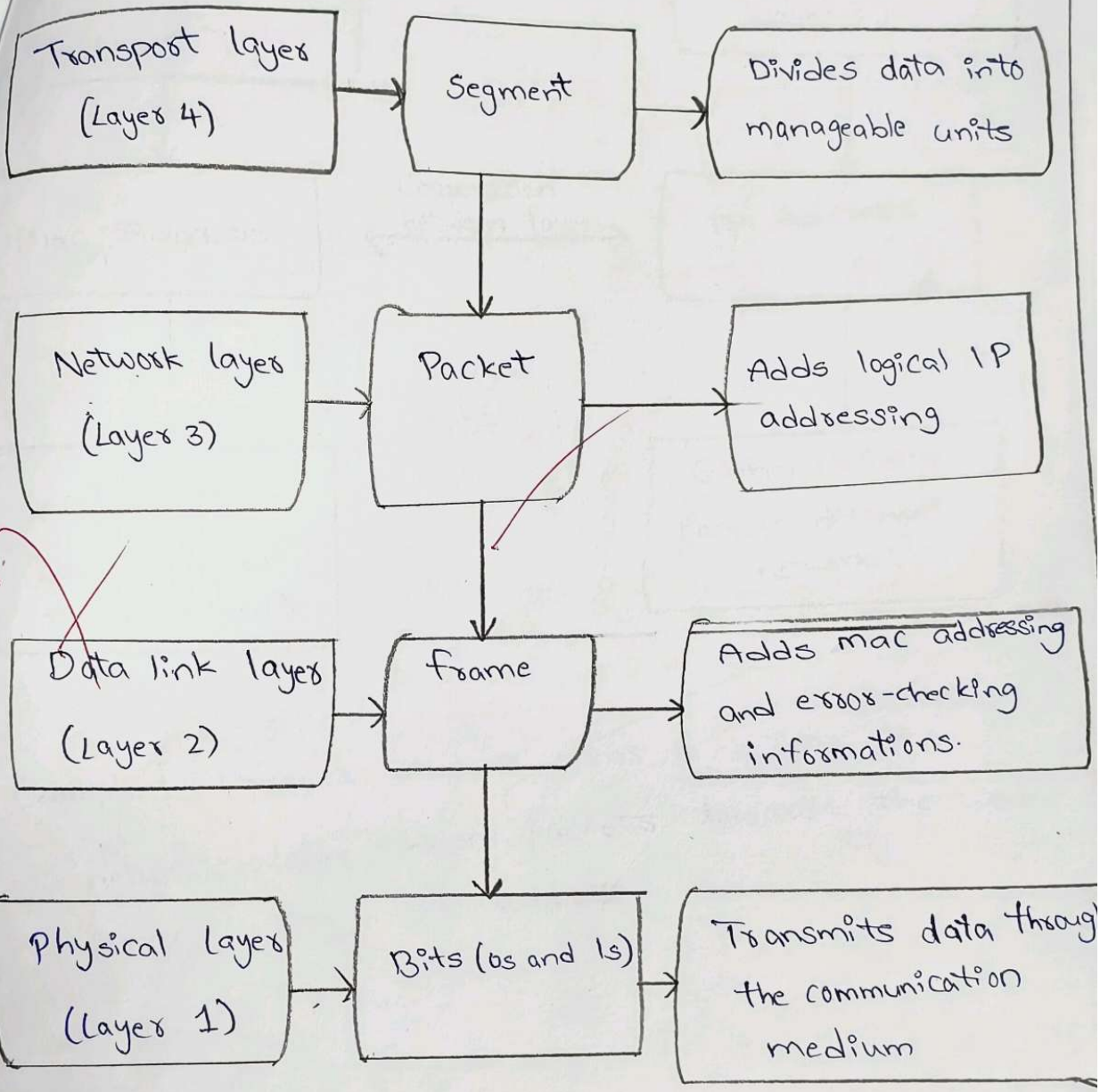
Q1. Identifying the missing OSI layer and explain their functions.

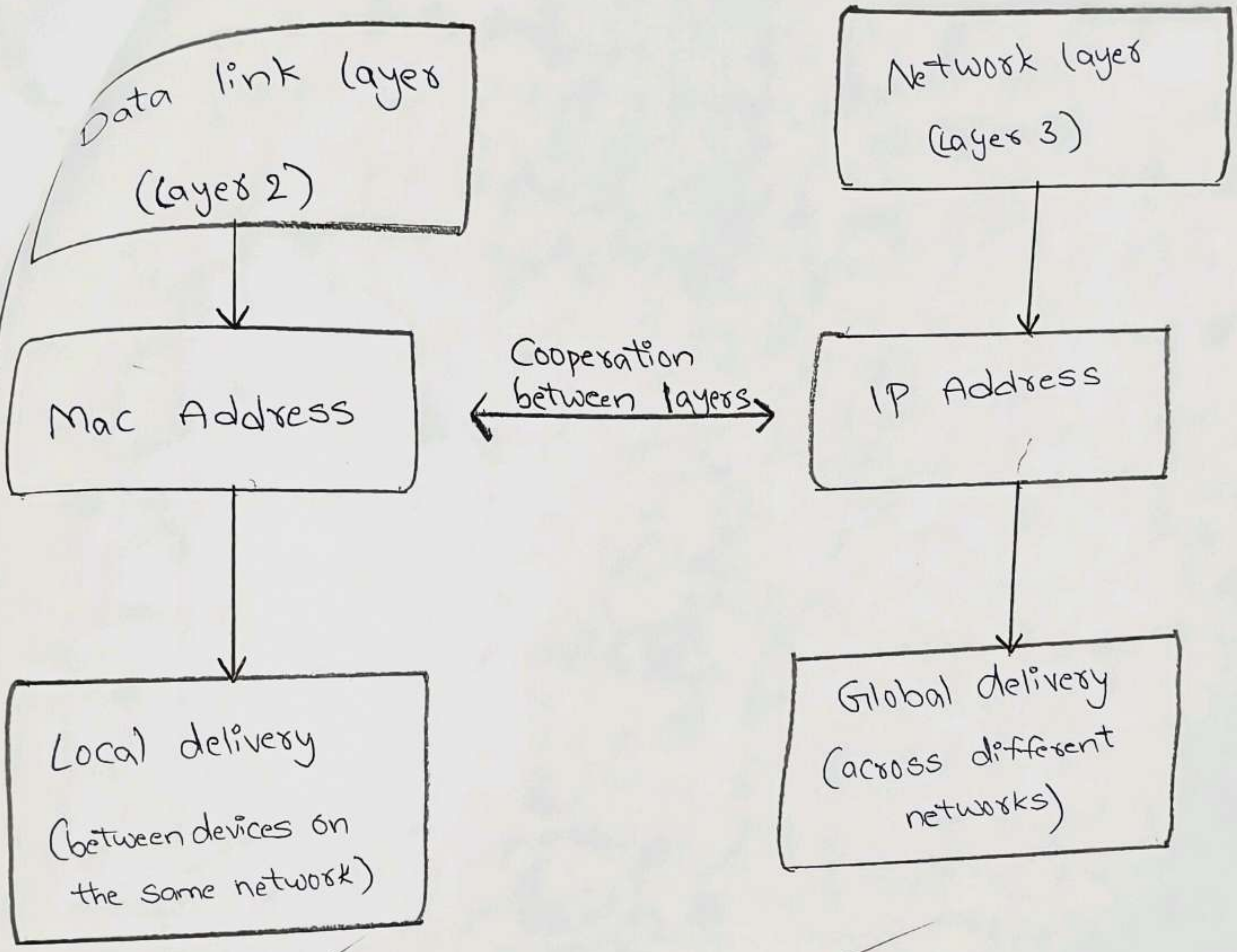
21/20




Shaik Abdul Adnan
(19722600084)

ata units and OSI layers :-






Example :- When a student opens a actions the network layer helps packets towards the web server using IP address.

# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

**CO1:** *Apply the functions of OSI layer in enabling reliable data transmission across networks.* (BL2, BL3)

*Assessment Tool Weightage: 30%*

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Annexure A

**TECHNICAL PRESENTATION**

*Code of Conduct:*

*I* Shaik Abdul Adnan (192260084) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

| Criteria | Weightage | Q1 30marks |
|---|---|---|
| Understanding of Concepts | 25 | 10 |
| Experimental Design | 30 | 10 |
| Use of Tools | 20 | 5 |
| Analysis of Results | 15 | 5 |
| Documentation | 10 | 0 |

**Total Marks** 25 **/30**

***Signature of the Faculty***

## TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

## RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable | Basic analysis with limited insights | Poor or incorrect analysis of results |

| | | | | | |
|---|---|---|---|---|---|
| | | | interpretatio n | Basic | Poor or incomplet |
| Documentati on | 10 | Clear, well-structured documentation with diagrams, code, and results | Organized documentati on with necessary details | Basic documentati on with missing details | Poor or incomplete documentati on |

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CSA0709

# Satellite Internet Communication Using OSI Layers

## Understanding Satellite Internet Through the Seven OSI Layers

**TECHNICAL PRESENTATION**
**CO 1 AT3**


Guided by:
Shankar R

**Presented by:**
**Shaik Abdul Adnan (1972260084)**

# Abstract

## Overview

**Satellite Internet Communication uses satellites to provide internet access to remote and underserved areas. Data travels between a user terminal, a satellite and a ground gateway before reaching the wider internet.**
**The OSI model explains this communication through seven layers, from physical radio transmission to application services. This mapping helps describe how data is transmitted, routed and delivered.**

# Introduction

## Why satellite internet uses layered communication

- Satellite internet connects user terminals to satellites that relay data to ground gateways and internet networks.
- Communication involves radio signals, link control, addressing, routing and application data.
- Weather, latency, signal strength and shared capacity can affect service quality.
- The OSI model separates communication into seven layers, making functions easier to explain and troubleshoot.
- Project goal: describe satellite internet by mapping its major functions to the OSI layers.

# Problem Statement

### Communication Path

Data travels from a user device through a satellite link and gateway to an internet server.

### Layered Network

The process combines radio transmission, link framing, IP addressing and application protocols.

### Performance Issues

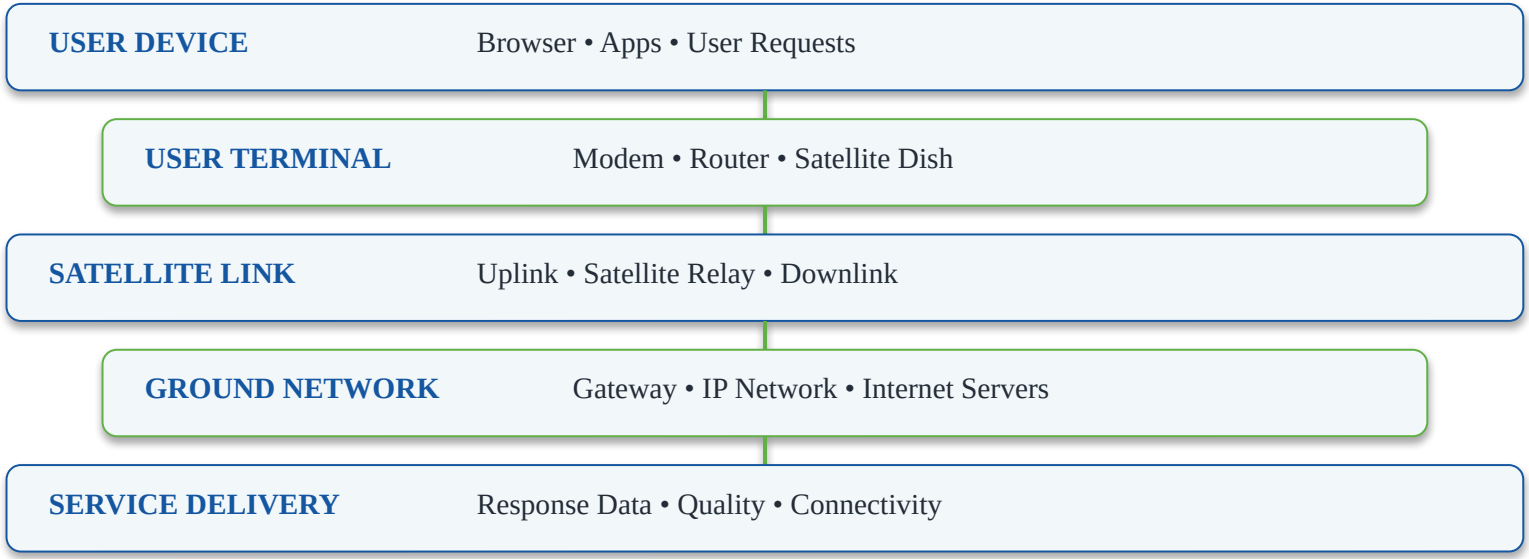
Long distance, weather and congestion can affect latency and reliability.

### Layered Explanation

A layer-by-layer view clarifies each function and supports troubleshooting.

**Proposed solution → explain satellite internet communication using the seven OSI layers.**

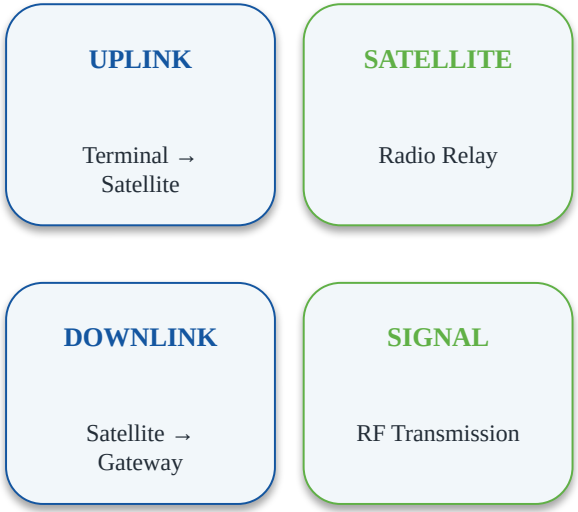
# System Architecture




Each layer performs a defined role, helping engineers isolate communication and connectivity issues.

# Layer 1 – Physical Layer

- Transmits raw bits as radio signals across the satellite link.
- Uses antennas, transceivers, frequency bands and modulation.
- Includes uplink from the user terminal and downlink from the satellite or gateway.
- Signal quality is affected by distance, interference, rain fade and antenna alignment.
- Defines the physical characteristics of the communication medium.



# Layers 2–4 – Link, Network and Transport

## Data Link Layer

Frames data, manages link access and helps detect transmission errors over a link.

## Network Layer

IP addressing and routing move packets between the gateway and destination networks.

## Transport Layer

TCP or UDP carries data between endpoints; TCP can retransmit lost data, while UDP reduces overhead.

**Communication flow**

**Frame → Route → Deliver**

# Layers 5–7 – Session, Presentation and Application

| | | |
|---|---|---|
| 1 | **Session Layer** | Coordinates communication sessions between applications where applicable. |
| 2 | **Presentation Layer** | Handles data representation, encoding and encryption functions when used by software. |
| 3 | **Application Layer** | Provides services such as web browsing, email, streaming and DNS to user applications. |
| 4 | **End-to-End Delivery** | Application data travels through lower layers and returns to the destination application. |

**Key idea: OSI is a reference model; real satellite systems implement protocols across these functions.**

# Future Enhancements

### Coverage Visualization

Map satellite footprints, gateways and user-terminal coverage areas.

### Link Performance

Compare latency, throughput, packet loss and availability under different conditions.

### Security Improvements

Explore authentication, encryption and secure network management.

### Resilience

Study backup links, failover and service continuity.

### Emerging Systems

Compare LEO, MEO and GEO satellite architectures and their trade-offs.

**Long-term vision → reliable, secure and accessible internet connectivity through satellite networks.**

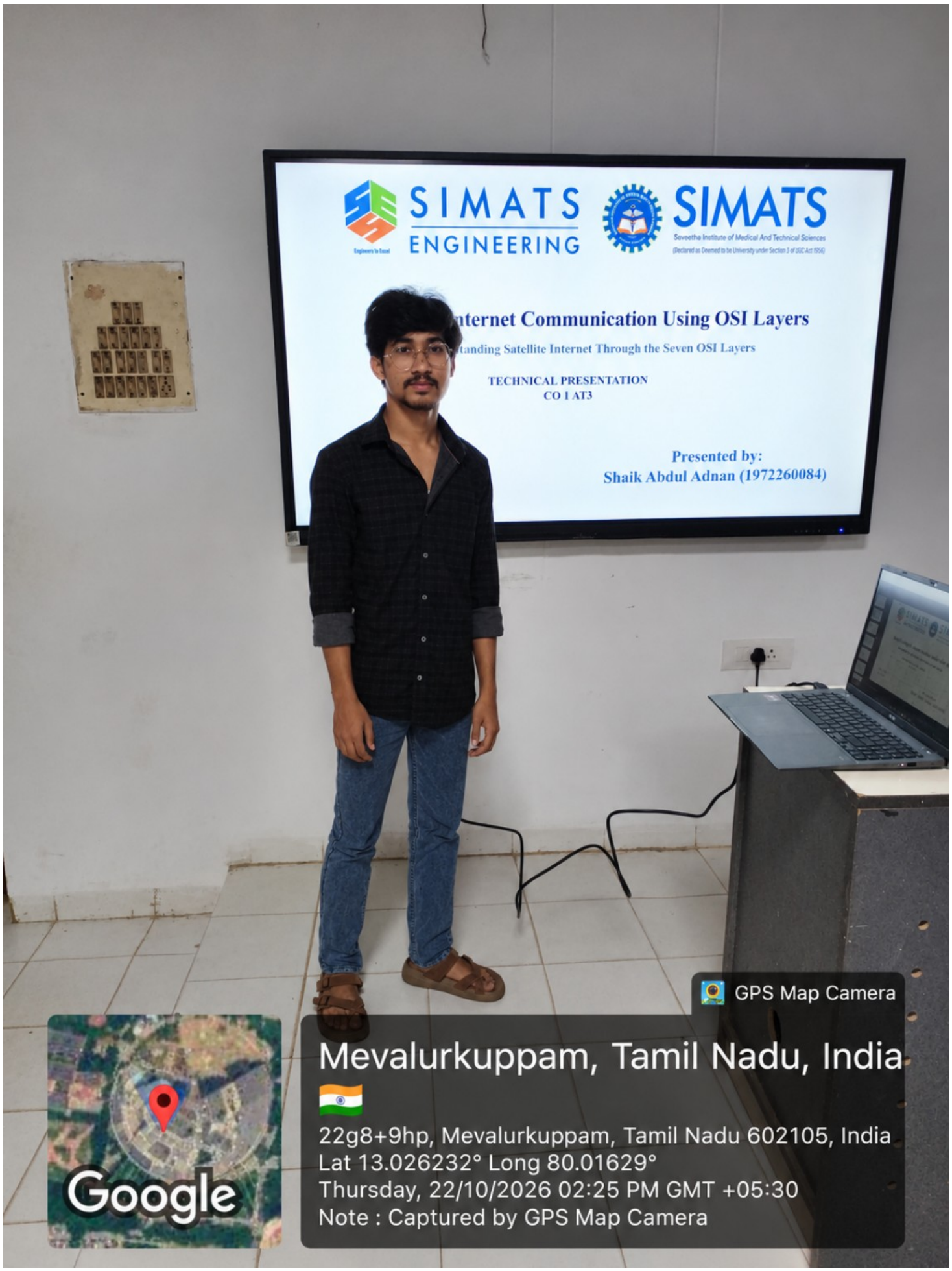
# Conclusion

Satellite Internet Communication provides internet connectivity to remote and rural areas where traditional networks are difficult to establish. The OSI model explains how data travels through the seven layers, from application services to physical transmission through satellites. Each layer performs a specific function, including data formatting, routing, addressing, and reliable communication. By understanding the OSI layers, we can better analyze, troubleshoot, and improve satellite internet communication systems. Overall, satellite communication plays an important role in providing global connectivity and supporting modern digital services.

# References


• International Telecommunication Union (ITU), satellite communications and radio regulations resources.
• A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5th ed., Pearson, 2011.
• J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, 8th ed., Pearson, 2021.
• ISO/IEC 7498-1, Information technology — Open Systems Interconnection — Basic Reference Model.
• NASA, educational resources on satellite communications and space communications systems.

Thank you


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Presented by:
Shaik Abdul Adnan (1972260084)
GPS Map Camera
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Thursday, 22/10/2026 02:25 PM GMT +05:30
Note : Captured by GPS Map Camera
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